

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	SWTID-2026-5841
Project Name	FitBuddy – AI Fitness Plan Generator
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the structure, readability and reusability of the FitBuddy codebase.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Four-space Python style used in all files.
2	Proper File Structure	Files and folders are logically organized	Yes	app/ (routes, schemas, database, gemini modules), templates/, static/, tests/, Dockerfile.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. user_id, workout_plan, nutrition_tip, current_plan, feedback.
4	Function / Method Names	Functions are descriptively named	Yes	generate_workout_gemini, update_workout_plan, upsert_user, save_updated_plan.
5	Code Comments	Inline and block comments explain logic	Partial	Docstrings and short comments on key functions; more can be added.
6	Modular Design	Code is split into reusable functions/modules	Yes	Routing, validation, AI calls and database access are separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Partial	Shared helpers avoid duplication; a small helper script (check_key.py) is kept separate.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	GeminiError, retry with backoff, 404/422/502 error pages, non-fatal tip failure.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	generate_text()	Python / google-gemini	Workout plan, plan update and nutrition tip	High
2	session_scope()	Python / SQLAlchemy	Every database read and write	High
3	UserInput and FeedbackRequest schemas	Python / Pydantic	Validation for both form routes	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
4	_render_result()	Python / Jinja2	Result page after generation and after feedback	Medium
5	FORMAT_RULES prompt text	Python	All workout prompts	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear separation of layers and files.
Readability	4	Short functions with clear names and type hints.
Reusability	4	Shared AI client, session helper and schemas.
Documentation / Comments	3	Docstrings on key functions; README present; more inline comments possible.
Overall Score	3.8	Team self-assessment out of 5.